

flea

NEWS **70** *June 2012*



<http://esanetworks.org/>

FLEA NEWS is a biannual newsletter about fleas (Siphonaptera). Recipients are urged to check any citations given here before including them in publications. Many of our sources are abstracting journals and current literature sources such as National Agricultural Library (NAL) Agricola, and National Library of Medicine (NLM) Medline, and citations have not necessarily been checked for accuracy or consistent formatting.

Recipients are urged to contribute items of interest to the profession for inclusion herein, including: Flea research citations from journals that are not indexed in Agricola or Medline databases, Announcements and Requests for material, Contact information for a Directory of Siphonapterists (name, mailing address, email address, and areas of interest - Systematics, Ecology, Control, etc.), Abstracts of research planned or in progress, Book reviews, Biography, Hypotheses and Literature Reviews, and Anecdotes. Send to:

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Editor, Flea News

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Editorial

Dear Flea News Reader,

Though most of the references listed here are about fleas (Insecta: Siphonaptera), some are about other ectoparasites, especially lice, mites, and ticks, or merely reference flea research. A few are about other fleas: flea beetles, flea markets, flea crustaceans. The reader should check all references.

ON TRAPPING HOSTS TO SURVEY PARASITES:

As a note to those of you who are trapping any hosts to collect fleas or other parasites, you should remember to record each of the hosts trapped that do NOT have fleas when you check it, along with the hosts that do. That sort of negative result might seem a wasted effort, but in fact, such "null host" data, which parasitologists often overlook in their intent to collect fleas and host records, are essential to calculate the population distribution and abundance of the flea species among the host population.

Other useful data to record are the number of traps in your trap-line, and your trapping effort ("trap-hours"), which can be used to estimate the population density of the host at that location. The density of the hosts varies, and this can affect the parasites.

R.L. Bossard, Editor, Flea News

Announcements

The 2012 Livestock Insect Workers Conference in Kalispell, Montana, USA was delightful and well-attended, with a record 48 presentations over a range of topics, and over 85 registered attendees, but nope, not a single flea paper. None at the annual meetings of the Entomological Society of America 2011, or American Society of Parasitologists 2012, either.

**

Ongoing Research Projects

Environmental study of vectors and reservoirs of zoonosis in Cape Verde.

In-progress Research, in cooperation with Cape Verde (M^a Soledad Gómez).

AECID (Agencia Española de Cooperación Internacional para el Desarrollo), Program for inter-university cooperation and scientific research (A103535611)).

Activities:

- Equipment of Cape Verde laboratories
- Training courses in the UNICV (Universidade de Cabo Verde)
- Design for sampling of the Culicidae of Santiago Island
- Mastozoological sampling on Santiago Island
- Taxonomic, faunistic, and systematic studies on bacteria, protozoa, helminthes, and arthropod parasites
- Treatment of samples, Data analysis, Reports.

**

Systematics and ecology of Siphonaptera ectoparasitic on Sigmodontine rodents from Northern Patagonia, Argentina.

Ongoing Doctoral Dissertation (J.P. Sanchez, Universidad de La Plata, Argentina; Asesor: M. Lareschi).

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Systematics, distribution, and ecology of Rhopalopsyllidae from Argentina, and their role as vectors of pathogens.

Ongoing Project (Titular: M. Lareschi).

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Life zones in the Great Basin Desert and biocenoses of coevolved mammals, fleas, and pathogens (Bossard, R.L. 2006. J. Parasitol. 92: 260-6).
Applicability of Island Biogeography Theory to flea species numbers on mammal hosts as islands.

Ongoing Projects (R.L. Bossard).

**

Entomology museums in California with flea collections and electronic databases:

Los Angeles Natural History Museum, Los Angeles, CA (Electronic database available)

Essig Museum of Entomology, Berkeley, CA, and the Smithsonian/Walter Reed (Electronic databases in progress)

Bohart Museum of Entomology, Davis, CA (Plans for electronic database)

California Academy of Sciences, San Francisco, CA (Distant future plans for electronic database)

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Jul 13, 2012

Hello All,

I just wanted to send out a reminder regarding the offer from Baylor [College of Medicine, Houston, Texas, USA] for genome sequencing our important veterinary arthropods. As a reminder, the genome is not the answer; rather it is a tool for finding more answers - similar to a computer and a microscope. In order for Baylor to choose some of our arthropods, we need to do some legwork, that is, find established colonies, and determine the approximate genome size.

From Baylor's site, for arthropods' genome sequencing, we recommend that the input DNA sequence be as non-polymorphic as possible. The ideal is a large haploid individual (for example from a large male hymenopteran) allowing the generation of up to 50 ug of genomic DNA. The next best is an inbred line, with 12-20 generations of sib-sib inbreeding. Twelve generations theoretically makes 90% of the genome homozygous, 20 is theoretically close to 99%, and any additional sib-sib inbreeding beyond this does not significantly reduce homozygosity in the sample. If inbreeding cannot be performed, the next best is a single large individual, so the assembler will only have to deal with a single diploid sequence. If the individual is small enough that multiple individuals are required for sufficient DNA, we recommend that the main library be made from a single individual using a low input DNA protocol, and DNA isolated from pooled individuals used for libraries of larger insert sizes requiring more DNA for gel cuts.

List of organisms with genomes already sequenced or in the works:

http://arthropodgenomes.org/wiki/Sequenced_genomes ; Baylor's current sequence:

<https://www.hgsc.bcm.edu/content/species-table> ; Baylor's role in the i5k arthropod genome project

<https://www.hgsc.bcm.edu/content/i5k-project> ; VectorBase (location of more genomes important to us): <http://www.vectorbase.org/> . Med/Vet arthropods with genomes started-

annotated (forgive my spelling): *Aedes aegypti*, *Anopheles darlingi/gambiae/funestus*, *Culex quinquefasciatus*, *Pediculus humanus*, *Rhodnius prolixus*, *Cimex lectularius* (transcriptome), *Cochliomyia hominivorax* (transcriptome), *Culex tarsalis*, *Glossina morsitans* (transcriptome), *Lutzomyia longipalpis/intermedia*, *Ochlerotatus triseriatus* (transcriptome), *Phlebotomus* species (transcriptome), *Simulium guianense/nigrimanum*, and *Ctenocephalides felis* (transcriptome). I

believe (and I may be wrong, not the first time) that the USDA labs are working on *Haematobia irritans* and *Stomoxys calcitrans* (?).

What I need from you is an email telling me: 1- The five arthropods you want to have genome sequenced, and the reason you selected each arthropod; 2- Information on an established colony of one of those arthropods that has been in colony for >20 generations.

I will compile everyone's thoughts and then send it around for verification before I send the information to Baylor. Here is my email: RFryxell@utk.edu .

Thanks,
Becky

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Knoxville, TN 37996-4560

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New Book Chapter

Ch. 7, *Yersinia pestis*: Plague.

A. Laudisoit, W. Ruppitsch, A. Stoeger, and A. Pietzka. 2012.
In Bsl3 and Bsl4 Agents: Epidemiology, Microbiology, and Practical Guidelines.
Mandy Elschner, Sally Cutler, Manfred Weidmann, and Patrick Butaye, (eds.), John Wiley & Sons, ISBN3527317155, 9783527317158, 400 pages.
<http://www.amazon.com/BSL3-BSL4-Agents-Mandy-Elschner/dp/3527317155> , accessed 14 July 2012.

“In one handy book, this reference gathers all the necessary information on 14 of the most commonly used dangerous groups of pathogens in biosafety level 3 and 4 laboratories. All the chapters are uniformly structured, with a brief overview of the microbiology, pathology, epidemiology and detection methods for each group. In addition, a whole chapter is devoted to the special biosafety requirements, disinfection, decontamination protocols, accident literature and accident procedures, as well as treatment options for all the organisms. This chapter is clearly marked and easy to find when opening the book. Essential literature for the increasing number of BSL3 or BSL4 labs worldwide and for medical facilities intervening in the case of an incident.”

Meetings

67th Annual International Conference on Diseases in Nature Communicable to Man (INCDNCM), July 22-24, 2012 - National Microbiology Laboratory, Humphrey Inn and Suites, Winnipeg, Canada. <http://sites.google.com/site/incdncm/2012-conference>

The 83rd Rocky Mountain Conference of Entomologists, August 5-8, 2012, Cathedral Ridge Conference and Retreat Center, Woodland Park, Colorado.

August 6-9, 2012 - Second Annual International Workshop on Malaria and Related Haemosporidian Parasites of Wildlife - National Conservation Training Center, Shepherdstown, West Virginia.

INTECOL 2013 / British Ecological Society Annual Meeting, Aug 18, 2013 - Aug 23, 2013, London, UK.

International Congress of Entomology, August 19-25, 2012, Daegu, South Korea.

Annual Meeting, Australian Mammal Society, Sep 23, 2012 - Sep 26, 2012, Port Augusta, South Australia.

2012 SOVE (Society for Vector Ecology) Annual Conference, St. Augustine, Florida, September 23-27 2012. dnorris@jhsph.edu

54th Peruvian Congress of Entomology, Peruvian Entomological Society (SEP), November 5-8, 2012, Cusco, Peru.

II Congreso Latinoamericano y las XXV Jornadas Argentinas de Mastozoología, Nov 6, 2012 - Nov 9, 2012, Joint Meeting of Argentinian Mammal Society (SAREM) and the II Latinamerican Congress, Buenos Aires, Argentina.

ESA Annual Meeting, November 11-14, 2012, Knoxville, Tennessee.

International Symposium on Insects, Entomological Society of Malaysia December 3-5, 2012, Kuala Lumpur. isoi.entoma@gmail.com

British Ecological Society Annual Meeting, Dec 18, 2012 - Dec 20, 2012, University of Birmingham, UK.

First Announcement
12th International Symposium of Ectoparasites in Pets (ISEP)

7th to 10th April 2013
Munich, Germany

Topics:
Arachno-Entomology
Emergence of Ectoparasites
Control of Ectoparasitic Vectors and Vector-borne Diseases
Efficacy and Safety of Ectoparasitocides
Resistance of Ectoparasites

www.isep2013.org

To be updated, please send an email (to isep@klifovet.com) or complete your contact details below to receive updated information.

Yes, please send me further information about the ISEP 2013 in Munich

Name: _____

Company/Institution: _____

Country: _____

Email: _____

Please return to:

Fax: +49 89 58 00 82 77 77

We are looking forward to welcome you in Munich in 2013!

Institute of Comparative KLIFOVET AG, Tropical Medicine and Parasitology, Munich,
Germany, Ludwig Maximilians- University, www.klifovet.com

Journals publishing papers on Siphonaptera from 1960 to 2012

There has been a linear increase in the number of flea papers published each year, from 86 in 1996 to 321 in 2011 (that is, an average of 16 papers more each year). The areas represented by these papers represent the diverse fields we would expect in connection with fleas: 34% of the these papers were in the Agricultural/Biological area, 16% in Immunology and Microbiology, 14% in Veterinary Science, 12% in Medicine, 9% in Environmental, 8% in Biochemistry, Genetics, and Molecular Biology, and the remaining 7% in other sciences. From 1960-present, the following major journals (monitored by Scopus database), published papers on "Siphonaptera". The name of journal is followed by the number of papers (in quotes), published during that time period in that journal. - Editor

SOURCE TITLE, "number of Siphonaptera papers"

Journal of Medical Entomology,"131"

Veterinary Parasitology,"84"
Zootaxa,"65"
Parasitology Research,"64"
Parasite,"54"
Journal of Parasitology,"48"
Journal of Vector Ecology,"46"
Vector Borne and Zoonotic Diseases,"45"
Parasitology,"44"
American Journal of Tropical Medicine and Hygiene,"44"
Proceedings of the Entomological Society of Washington,"42"
Journal of Economic Entomology,"37"
Medical and Veterinary Entomology,"37"
Molecular Phylogenetics and Evolution,"33"
Journal of Wildlife Diseases,"31"
Emerging Infectious Diseases,"31"
Parazitologiya,"30"
Veterinary Dermatology,"29"
Memorias do Instituto Oswaldo Cruz,"28"
Biological Control,"25"
Journal of Experimental Biology,"21"
Journal of Feline Medicine and Surgery,"21"
Chinese Journal of Endemiology,"21"
Veterinary Therapeutics,"20"
Annals of the New York Academy of Sciences,"20"
Revue De Medecine Veterinaire,"20"
Molecular Ecology,"19"
Oecologia,"19"
Plos One,"19"
Journal of the American Veterinary Medical Association,"18"
Veterinary Record,"18"
Journal of Animal Ecology,"18"
Oikos,"18"
Journal of Biogeography,"17"
International Journal for Parasitology,"17"
Proceedings of the Royal Society B Biological Sciences,"17"
Entomological Review,"17"
Applied and Environmental Microbiology,"16"
Comparative Immunology Microbiology and Infectious Diseases,"16"
Journal of Clinical Microbiology,"16"
Journal of Evolutionary Biology,"15"
Pest Management Science,"15"
Ecography,"15"
Coleopterists Bulletin,"15"
Parasites and Vectors,"15"

Insect Biochemistry and Molecular Biology,"15"
 Veterinary Research,"15"
 Biologia,"14"
 Entomologia Experimentalis Et Applicata,"14"
 Veterinary Microbiology,"14"
 Zoologica Scripta,"14"
 Canadian Journal of Zoology,"14"
 Annals of Carnegie Museum,"13"
 Cladistics,"13"
 Functional Ecology,"13"
 Behavioral Ecology,"13"
 Annual Review of Entomology,"13"
 Trends in Parasitology,"13"
 Canadian Entomologist,"12"
 Comparative Parasitology,"12"
 American Journal of Veterinary Research,"12"
 Parazitologia,"12"
 Acta Parasitologica,"11"
 Journal of Insect Physiology,"11"
 Systematic Entomology,"11"
 European Journal of Entomology,"11"
 Ecology,"11"
 Annals of the Entomological Society of America,"11"
 Veterinary Immunology and Immunopathology,"10"
 Ecological Entomology,"10"
 Veterinary Medicine,"10"
 Veterinary Clinics of North America Small Animal Practice,"10"
 Journal of Agricultural and Urban Entomology,"10"
 Kleintierpraxis,"10"
 Australian Journal of Entomology,"9"
 Proceedings of the National Academy of Sciences of the United States of America,"9"
 Experimental and Applied Acarology,"9"
 Journal of Chemical Ecology,"9"
 Environmental Entomology,"9"
 American Naturalist,"9"
 Zoological Journal of the Linnean Society,"9"
 Nature,"8"
 Biological Journal of the Linnean Society,"8"
 Crop Protection,"8"
 Clinical Microbiology and Infection,"8"
 Pestology,"8"
 Animal Behaviour,"8"
 Entomological News,"8"
 Advances in Experimental Medicine and Biology,"7"

Environmental Health Perspectives,"7"
Veterinary Technician,"7"
Insect Molecular Biology,"7"
Environmental Toxicology and Chemistry,"7"
Point Veterinaire,"7"
Biologia Section Zoology,"7"
Wildlife Research,"7"
Veterinary Economics,"7"
Neotropical Entomology,"7"
Western North American Naturalist,"7"
Invertebrate Systematics,"7"
Journal of Bacteriology,"7"
Southwestern Naturalist,"7"
Chemosphere,"7"
Journal of the South African Veterinary Association,"7"
Southeast Asian Journal of Tropical Medicine and Public Health,"6"
Journal of Applied Entomology,"6"
Onderstepoort Journal of Veterinary Research,"6"
Compendium Continuing Education for Veterinarians,"6"
Journal of Entomological Science,"6"
Journal of the American Mosquito Control Association,"6"
Acta Ornithologica,"6"
Evolutionary Ecology,"6"
Mastozoologia Neotropical,"6"
Annals of Tropical Medicine and Parasitology,"6"
Biocontrol,"6"
Journal of Natural History,"6"
Zoologicheskii Zhurnal,"6"
Phytoparasitica,"6"
Microbiology,"6"
Rangeland Ecology and Management,"6"
Infection and Immunity,"6"
Pesticide Biochemistry and Physiology,"6"
Biology Bulletin,"6"
Infection Genetics and Evolution,"6"
Ecology Letters,"6"
Journal of Insect Science,"6"
Molecular Microbiology,"6"
Revista Brasileira De Parasitologia Veterinaria,"6"
Journal of Zoo and Wildlife Medicine,"6"
Arthropod Structure and Development,"6"
Mammalia,"6"
Irish Veterinary Journal,"6"
Parasitology Today,"6"

Acta Zoologica Academiae Scientiarum Hungaricae,"6"
 In Practice,"6"
 Acta Zoologica Lituanica,"6"
 Zoonoses and Public Health,"5"
 Micron,"5"
 Great Lakes Entomologist,"5"
 Australian Veterinary Practitioner,"5"
 Hydrobiologia,"5"
 Ibis,"5"
 Journal of Insect Behavior,"5"
 Preventive Veterinary Medicine,"5"
 Cutis,"5"
 Pratique Medicale Et Chirurgicale De L Animal De Compagnie,"5"
 Journal of the Kansas Entomological Society,"5"
 Archives of Environmental Contamination and Toxicology,"5"
 Journal of the New York Entomological Society,"5"
 PloS Neglected Tropical Diseases,"5"
 Bulletin of the American Museum of Natural History,"5"
 Florida Entomologist,"5"
 Global Ecology and Biogeography,"5"
 Biodiversity and Conservation,"5"
 Systematic Biology,"5"
 Annales De La Societe Entomologique De France,"5"
 Transactions of the Royal Society of Tropical Medicine and Hygiene,"5"
 Genome,"5"
 Journal of Zoological Systematics and Evolutionary Research,"5"
 Australian Veterinary Journal,"5".

Featured research

Beaucournu J.-C., Meheretu J., Welegerima K., Mergey T. & Laudisoit A. 2012. Description de *Nosopsyllus (N.) atsbi* n. sp. (Siphonaptera : Ceratophyllidae) d' Ethiopie et révision de l'espèce affine *N. (N.) incisus* (Jordan & Rothschild, 1913); discussion biogéographique. Parasite 19, 31-40.

Beaucournu J.-C., Munoz-Leal S.A. & Gonzalez-Acuna D. 2012. *Dasypsyllus patagonicus* n. sp. de l'extrême sud du Chili. Bulletin de la Société entomologique de France, 117, 119-122.

Dasypsyllus patagonicus n. sp. from Southern Chile (Siphonaptera, Ceratophyllidae). A new flea belonging to the genus *Dasypsyllus* Baker, 1905, *D. patagonicus* n. sp., is described from a single

female from Southern Chile. The features shown by this specimen differ from all other taxa previously described and are particular enough to consider the definition of a new sub-genus when the male is known.

Laudisoit A., Prié V & Beaucournu J.-C.- Note sur les Ischnopsyllinae de la région afrotropicale-V: Une nouvelle espèce de *Lagaropsylla* Jordan & Rothschild, 1921 de Madagascar (Insecta-Siphonaptera-Ischnopsyllidae). *Zoosystema*, 2012, à paraître.

Beaucournu J.-C., Mergey T., Munoz- Léal S.A. & González-Acuña D.- Description de *Tunga bonneti* n. sp. (Insecta-Siphonaptera-Tungidae) et notes sur sa spécificité, sa chorologie et sa phénologie. *Parasite*, à paraître.

Lemaire J.-M. & Beaucournu J.-C.- Une localité nouvelle et un hôte nouveau pour la puce d'Oiseau *Ceratophyllus vagabundus alpestris* Jordan, 1926 (Siphonaptera, Ceratophyllidae). *Biocosme*, à paraître.

Beaucournu J.-C., Degeilh B., Mergey T., Munoz-Leal S.A. & González-Acuña D.- Notes sur le genre *Tunga* Jarocki, 1838 (Insecta-Siphonaptera-Tungidae). 1ère partie : taxonomie et répartition, phylogénie, eco-éthologie, rôle pathogène. *Parasite*, soumis.

Beaucournu J.-C. & Laudisoit A.- Contribution à l'étude de *Ctenophthalmus* (*Ethioctenophthalmus*) *evidens* (Siphonaptera – Ctenophthalmidae) et description de 2 sous-espèces nouvelles. *Bulletin de la Société entomologique de France*, soumis.

Perrichot V., Beaucournu J.-C. & Velten J.- First extinct genus of a flea (Siphonaptera - Pulicidae) in Miocene Amber from the Dominican Republic. *Zootaxa*, à paraître 2012 (*Eospilopsyllus kobberti* n. gen. et n. sp.).

**

Chouikha, I., Hinnebusch, B.J. 2012. *Yersinia*-flea interactions and the evolution of the arthropod-borne transmission route of plague. *Current Opinion in Microbiology* Article in Press.

Yersinia pestis, the causative agent of plague, is unique among the enteric group of Gram-negative bacteria in relying on a blood-feeding insect for transmission. The *Yersinia*-flea interactions that enable plague transmission cycles have had profound historical consequences as manifested by human plague pandemics. The arthropod-borne transmission route was a radical ecologic change from the food-borne and water-borne transmission route of *Yersinia pseudotuberculosis*, from which *Y. pestis* diverged only within the last 20 000 years. Thus, the interactions of *Y. pestis* with its flea vector that lead to colonization and successful transmission are the result of a recent evolutionary adaptation that required relatively few genetic changes. These changes from the *Y. pseudotuberculosis* progenitor included loss of insecticidal activity,

increased resistance to antibacterial factors in the flea midgut, and extending *Yersinia* biofilm-forming ability to the flea host environment.

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Czenze, Z. J. and H. G. Broders. 2011. Ectoparasite Community Structure of Two Bats (*Myotis lucifugus* and *M. septentrionalis*) from the Maritimes of Canada. Journal of Parasitology Research 2011.

Prevalence of bat ectoparasites on sympatric *Myotis lucifugus* and *M. septentrionalis* was quantitatively characterized in Nova Scotia and New Brunswick by making systematic collections at swarming sites. Six species of ectoparasite were recorded, including *Myodopsylla insignis*, *Spinturnix americanus*, *Cimex adjunctus*, *Macronyssus crosbyi*, *Androlaelap scasalis*, and an unknown species of the genus *Acanthophthirius*. Male *M. lucifugus* and *M. septentrionalis* had similar prevalence of any ectoparasite (22% and 23%, resp.). Female *M. lucifugus* and *M. septentrionalis* had 2-3 times higher prevalence than did conspecific males (68% and 44%, resp.). Prevalence of infection of both genders of young of the year was not different from one another and the highest prevalence of any ectoparasite (*M. lucifugus* 64%, *M. septentrionalis* 72%) among all bat groups. Ectoparasite prevalence and intensity varied positively with roost group size and negatively with grooming efficacy and energy budgets, suggesting that these variables may be important in ectoparasite community structure.

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Eisen, R. J., J. N. Borchert, J. T. Mpanga, L. A. Atiku, K. MacMillan, K. A. Boegler, J. A. Montenieri, A. Monaghan, and K. L. Gage. 2012. Flea Diversity as an Element for Persistence of Plague Bacteria in an East African Plague Focus. PLoS ONE 7 (4).

Plague is a flea-borne rodent-associated zoonotic disease that is caused by *Yersinia pestis* and characterized by long quiescent periods punctuated by rapidly spreading epidemics and epizootics. How plague bacteria persist during inter-epizootic periods is poorly understood, yet is important for predicting when and where epizootics are likely to occur and for designing interventions aimed at local elimination of the pathogen. Existing hypotheses of how *Y. pestis* is maintained within plague foci typically center on host abundance or diversity, but little attention has been paid to the importance of flea diversity in enzootic maintenance. Our study compares host and flea abundance and diversity along an elevation gradient that spans from low elevation sites outside of a plague focus in the West Nile region of Uganda (~725-1160 m) to higher elevation sites within the focus (~1380-1630 m). Based on a year of sampling, we showed that host abundance and diversity, as well as total flea abundance on hosts was similar between sites inside compared with outside the plague focus. By contrast, flea diversity was significantly higher inside the focus than outside. Our study highlights the importance of considering flea diversity in models of *Y. pestis* persistence.

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Maximilian Fraulob, Benjamin Wipfler, Frank Hünefeld, Hans Pohl, Rolf G. Beutel. 2011. The larval abdomen of the enigmatic Nannochoristidae (Mecoptera, Insecta). *Arthropod Structure and Development*, Volume 41, Issue 2, March 2012, Pages 187-198, ISSN 1467-8039, 10.1016/j.asd.2011.11.001. <http://www.sciencedirect.com/science/article/pii/S1467803911001149>)

External and internal structures of the larval abdomen of Nannochorista are described in detail, with emphasis on the posterior segments. The results are compared with conditions found in other groups of Antliophora, especially the mecopteran subgroups Boreidae and Pistillifera. Like the entire postcephalic body, the larval abdomen of Nannochorista is extremely slender and nearly cylindrical. The anterior segments are largely unmodified. The surface is smooth and lacks any protuberances or prolegs. The term “cloaca” for the posterior membranous pouch of Nannochorista sp. is morphologically unjustified. A list of muscles of segments IX and X is presented. The abdominal musculature was partly homologized following Snodgrass. The muscles of segment X are highly modified. They move the membranous pouch, the anal papillae, and the terminal lobes. The presence of these structures is likely an adaptation to the specific aquatic life style of nannochoristid larvae. The anal papillae are possibly homologous to the 4-lobed terminal attachment apparatus of larvae of Caurinus (Boreidae) and Pistillifera (Panorpidae, Bittacidae, Choristidae) but this is uncertain. The specific condition in both groups, i.e. two retractile papillae with tracheae and Malpighian tubules in Nannochoristidae, and a 4-lobed exposed attachment device in Pistillifera + Boreidae (groundplan) are very likely autapomorphic for both groups, respectively. A slender abdomen with smooth surface is very likely plesiomorphic within Antliophora and Mecopterida. This condition is found in Trichoptera (partim), Nannochoristidae, Siphonaptera, and many basal groups of Diptera. An eruciform or scarabaeiform body shape with a soft, largely unsclerotised cuticle is probably a synapomorphy of Boreidae and Pistillifera. The presence of ventral protuberances resembling prolegs on the anterior segments is an autapomorphy of the latter group. The homology of paired or unpaired terminal appendages of segment X is uncertain. However, the specific condition of paired and 3-segmented appendages with hooks in Nannochoristidae is almost certainly autapomorphic for this family. The protracted opening of the hind gut on the membranous pouch is another potential autapomorphy of Nannochoristidae. Aquatic habits of larvae, also very likely an apomorphic condition, have likely evolved several times independently in Antliophora.

**

Carlos Feliu, Mercedes López, María S. Gómez, Jordi Torres, Santiago Sánchez, Jordi Miquel, Néstor Abreu-Acosta, Juan M. Segovia, Aarón Martín-Alonso, Isabel Montoliu, Mercedes Villa, Ángela Fernández-Álvarez, Abdoulaye J.S. Bakhoun, Basilio Valladares, Jorge Orós, and Pilar Foronda. 2012. Parasite fauna of rodents (Murinae) from El Hierro (Canary Islands, Spain): a multidisciplinary approach. *Acta Parasitologica*, 57(2), 171–178.

The parasite fauna (protozoa, helminths and insects) of the two most widespread Murinae rodents in El Hierro (Canary Islands, Spain), the black rat (*Rattus rattus*) and the house mouse (*Mus musculus domesticus*) was studied. Faunistic, ecological, ecotoxicological data, as well as

information on the biology of some nematode parasites of *R. rattus* are provided. The present work is unprecedented in the Canary Islands, and provides the first data on the parasite biodiversity in Murinae from the archipelago. Concerning to parasitofaunas stands out: a) impoverishment of biodiversity of helminths respect of which have the same hosts in other islands; b) increasing the number of species of Siphonaptera, even compared with flea species that parasitize the same hosts from continental biotopes.

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Vicente, S. S. and M. S. Gómez López. 2012. Presence of *Stenoponia tripectinata* (Tiraboschi, 1902) (Siphonaptera, Ctenophthalmidae) in Murine (Rodentia) from the Canary Islands. *Acta Parasitologica* 57 (2): 190-193.

This article reports the geographical and host distribution of the flea *Stenoponia tripectinata* on the Canary Islands. *S. tripectinata* is widely distributed throughout the Mediterranean and North Africa as a parasite of Muridae rodents. To date, Gran Canaria is the only island of the archipelago where *S. tripectinata* had been found. In this report, *S. tripectinata* has appeared parasitizing 116 specimens of *Mus musculus* out of a total of 660, and only 2 *Rattus rattus* of 215 captured. All the trapped Muridae hosts found to be parasitized by *S. tripectinata* came from humid biotopes. The results showed that *S. tripectinata* is present on all the western Canary Islands and on one of the eastern islands, Gran Canaria, the only island already reported. The detection of *S. tripectinata* on El Hierro, La Palma, La Gomera and Tenerife represents the first records of this flea species on those Canary Islands.

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Fontaine B, van Achterberg K, Alonso-Zarazaga MA, Araujo R, Asche M, et al. 2012. New Species in the Old World: Europe as a Frontier in Biodiversity Exploration, a Test Bed for 21st Century Taxonomy. *PLoS ONE* 7(5): e36881. doi:10.1371/journal.pone.0036881 [open access, freely available online]

The number of described species on the planet is about 1.9 million, with ca. 17,000 new species described annually, mostly from the tropics. However, taxonomy is usually described as a science in crisis, lacking manpower and funding, a politically acknowledged problem known as the Taxonomic Impediment. Using data from the Fauna Europaea database and the Zoological Record, we show that contrary to general belief, developed and heavily-studied parts of the world are important reservoirs of unknown species. In Europe, new species of multicellular terrestrial and freshwater animals are being discovered and named at an unprecedented rate: since the 1950s, more than 770 new species are on average described each year from Europe, which add to the 125,000 terrestrial and freshwater multicellular species already known in this region. There is no sign of having reached a plateau that would allow for the assessment of the magnitude of European biodiversity. More remarkably, over 60% of these new species are described by non-professional taxonomists. Amateurs are recognized as an essential part of the workforce in

ecology and astronomy, but the magnitude of non-professional taxonomist contributions to alpha-taxonomy has not been fully realized until now. Our results stress the importance of developing a system that better supports and guides this formidable workforce, as we seek to overcome the Taxonomic Impediment and speed up the process of describing the planetary biodiversity before it is too late.

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Gao, T. -P, C. -K Shih, X. Xu, S. Wang, and D. Ren. 2012. Mid-Mesozoic Flea-Like Ectoparasites of Feathered Or Haired Vertebrates. *Current Biology* 22 (8): 732-735. [edited summary]

Parasite-host associations among insects and mammals or birds are well attended by neontological studies. An Eocene bird louse compression fossil and several flea specimens from Eocene and Oligocene ambers and, reported to date, are exceptionally similar to living louse and flea taxa. But the origin, morphology, and early evolution of parasites and their associations with hosts are poorly known and due to sparse records of putative ectoparasites with uncertain classification in the Mesozoic, most lacking mouthpart information and other critical details of the head morphology. Here we present two primitive flea-like species assigned to the Pseudopulicidae Gao, Shih et Ren familia nova (fam. nov.), *Pseudopulex jurassicus* Gao, Shih et Ren genus novum et species nova (gen. et sp. nov) from the Middle Jurassic and *P. magnus* Gao, Shih et Ren sp. nov. from the Early Cretaceous in China. They exhibit many features of ectoparasitic insects. Large body size and long serrated stylets for piercing tough and thick skin or hides of hosts suggest that these primitive ectoparasites might have lived on and sucked the blood of relatively large hosts, such as contemporaneous feathered dinosaurs and/or pterosaurs or medium-sized mammals (found in the Early Cretaceous, but not the Middle Jurassic).

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Huang, Diying, Michael S. Engel, Chenyang Cai, Hao Wu & André Nel. 2012. Diverse transitional giant fleas from the Mesozoic era of China. *Nature* 483: 201–204.

Fleas are one of the major lineages of ectoparasitic insects and are now highly specialized for feeding on the blood of birds or mammals. This has isolated them among holometabolan insect orders, although they derive from the Antliophora (scorpionflies and true flies). Like most ectoparasitic lineages, their fossil record is meagre and confined to Cenozoic-era representatives of modern families, so that we lack evidence of the origins of fleas in the Mesozoic era. The origins of the first recognized Cretaceous stem-group flea, *Tarwinia*, remains highly controversial. Here we report fossils of the oldest definitive fleas—giant forms from the Middle Jurassic and Early Cretaceous periods of China. They exhibit many defining features of fleas but retain primitive traits such as non-jumping hindlegs. More importantly, all have stout and elongate sucking siphons for piercing the hides of their hosts, implying that these fleas may be rooted among the pollinating ‘long siphonate’ scorpionflies of the Mesozoic. Their special morphology suggests that their earliest hosts were hairy or feathered ‘reptilians’, and that they radiated to mammalian and bird hosts later in the Cenozoic.

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Hastriter, Michael. 2009. A description of four new species of fleas (Insecta, Siphonaptera) from Angola, Ethiopia, Papua New Guinea, and Peru. *ZooKeys* 8 : 39-61.

Four new species of fleas are described: *Aphropsylla truncata* sp. n. (Ethiopia), *Ectinorus hirsutus* sp. n. (Peru), *Rhinolophopsylla traubi* sp. n. (Angola), and *Thaumapsylla wilsoni* sp. n. (Papua New Guinea). Our understanding of the genus *Aphropsylla* is deficient, therefore a discussion of host/parasite relationships and new records from Uganda are provided. A key is provided for the genus *Aphropsylla*, while representatives of the other three genera are associated with existing keys to facilitate their identification. The presence of lucodiscs on *Aphropsylla* and other genera is briefly discussed. The occurrence of lucodiscs among representatives of the order Siphonaptera deserves further investigation.

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Kiefer, D. , Dalantai, G., Damdindorj, T., Riehm, J.M., Tomaso, H., Zöller, L., Dashdavaa, O., (...), Scholz, H.C. 2012. Phenotypical characterization of Mongolian *Yersinia pestis* strains. *Vector-Borne and Zoonotic Diseases* 12 (3) , pp. 183-188.

Although Mongolia is regarded as one of the possible places of plague radiation, only few data are available from Mongolian *Yersinia pestis* strains. In this study a total of 100 Mongolian *Y. pestis* strains isolated from wild mammals and their parasites between the years 1960 and 2007 were analyzed for their phenotype. All strains grew well on selective Cefsulodin-Irgasan-Novobiocin agar and were positive for the F1-antigen, the F1-gene (*caf1*), and the plasminogen activator gene (*pla*). Biochemical analyses using the API20E- system identified 93% of the strains correctly as *Y. pestis*. The BWY in-house system consisting of 38 biochemical reactions was used to differentiate among *Y. pestis* subspecies *pestis* biovars *Antiqua* and *Medievalis* and also between the subspecies *microtus* biovars *Ulegeica* and *Caucasica*. Antibiotic susceptibility testing according to Clinical and Laboratory Standards Institute-guidelines identified one strain as being multiresistant. This strain was isolated from a wildlife rodent with no anthropogenic influence and thus suggests naturally acquired resistance.

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Kimble-Evans, A. 2011. Poisoned Pollen: Research shows pesticide is responsible for decrease in honeybee population. *Organic gardening*, 58(5):72.

“New research from the USDA Bee Research Laboratory and Pennsylvania State University shows that a toxic pesticide, imidacloprid, weakens bees and makes them more susceptible to the parasitic fungus *Nosema*, even when the pesticide is present at extremely low levels. Imidacloprid is one of a new class of pesticides called neonicotinoids whose use in the United States has grown in the past decade. It is commonly used for insect control on food crops and

lawns, for flea control on pets, and indoors to kill household pests. As a systemic insecticide, it is absorbed by plants and makes leaves and other plant parts toxic to feeding insects.

“It is a new threat,” says Jeff Pettis, entomologist at the USDA Bee Research Laboratory. “Ten years ago, almost all pesticides were sprayed or dusted topically. Bees only came in contact with drift when they landed on a plant. This new class of pesticide can be expressed in the nectar and the pollen.” While symptoms of neonicotinoid poisoning in honeybees look strikingly similar to CCD (memory loss, navigation disruption, paralysis, and death), Pettis says it is a mistake to lay blame on a single cause. Researchers continue to study the complex interactions that could be causing the bee die-offs.”

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Kuznetsov, A.; Osipov, V.; Knyazeva, T.; Sintsov, V.; Matrosov, A. 2010. Distribution of fleas in rodent colonies in the Northwestern Precaspian region. *Entomological Review* 90(8): 1014-1023.

Quantitative characteristics of phoresy of fleas on their rodent hosts inhabiting the sand landscapes of the northwestern Precaspian region were obtained. The structure of parasitic contact networks formed in the process was determined. The probability of a host change by fleas as the result of periodic feeding was assessed as high. The parameters of flea distribution were shown to be similar in the Northwestern Precaspian region and the Volgo-Ural sands. The ability of the flea *Xenopsylla conformis* to parasitize on social voles without significantly decreasing its abundance was revealed.

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Lareschi M, Autino, MM Diaz, R Barquez. 2011. Taxonomy and Distribution of *Nonnapsylla* Wagner, 1938 (Siphonaptera, Stephanocircidae, Craneopsyllinae). *Journal of Parasitology* 97 (5): 954-955. [listed in *Flea News* 69 December 2011]

Sanchez J., V. Amor, E. A. Bazán-León, R.A. Vásquez & M. Lareschi. 2012. Redescription of *Neotyphloceras chilensis* Jordan, new status (Siphonaptera: Ctenophthalmidae: Neotyphloceratini). *Zootaxa* (3259): 51-57.

Ezquiaga M.C. & M. Lareschi. 2012. Ultrastructure of the Eggs of *Malacopsylla grossiventris* and *Phthiropsylla agenoris* (Siphonaptera, Malacopsyllidae). *Journal of Parasitology* (in press) doi: <http://dx.doi.org/10.1645/GE-3062.1>

Sanchez J.P. & M. Lareschi. 2013. The fleas (Insecta: Siphonaptera) parasites of sigmodontine rodents (Cricetidae) from Northern Patagonia, Argentina. *Comparative Parasitology* 80 (1) (in press).

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Lockwood, J.A. 2012. Insects as weapons of war, terror, and torture. *Annual Review of Entomology* 57, pp. 205-227.

For thousands of years insects have been incorporated into human conflict, with the goals of inflicting pain, destroying food, and transmitting pathogens. Early methods used insects as "found" weapons, functioning as tactical arms (e.g., hurled nests) or in strategic habitats (e.g., mosquito-infested swamps). In the twentieth century the relationship between insects and disease was exploited; vectors were mass-produced to efficiently deliver pathogens to an enemy. The two most sophisticated programs were those of the Japanese in World War II with plague-infected fleas and cholera-coated flies and of the Americans during the Cold War with yellow fever-infected mosquitoes. With continued advances, defenses in the form of insecticides and vaccines meant that insects were no longer considered as battlefield weapons. However, in recent times sociopolitical changes have put insects back into the realm of human conflict through asymmetrical conflicts pitting combatants from non-industrialized regions against forces from militarily and economically superior nations.

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Malkova, M. G. and A. K. Tantsev. 2011. Zonal Types of Host-Parasite Complexes of Arthropods and Small Mammals in the West Siberian Plain. *Entomological Review* 91 (9): 1196-1201.

Comparative analysis of the composition, population structure, and landscape distribution of small mammals and associated parasitic arthropods was performed for the plain part of West Siberia. Four main zonal complexes were distinguished, corresponding to different landscape zones or subzones: tundra, forest, forest-steppe (the northern forest-steppe subzone), and steppe (the southern forest-steppe subzone and steppes). The parasite specificity of each complex is defined by different systematic and ecological groups of arthropods: the tundra complex is defined by epizootic gamasid mites (Acari: Parasitiformes: Mesostigmata), the forest and forest-steppe complexes are defined by ticks (Acari: Parasitiformes: Ixodidae) and fleas (Insecta: Siphonaptera), and the steppe complex is defined by fleas and nidicolous gamasid mites.

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Reijniers, J., S. Davis, M. Begon, J. A. P. Heesterbeek, V. S. Ageyev, and H. Leirs. 2012. A Curve of Thresholds Governs Plague Epizootics in Central Asia. *Ecology Letters* 15 (6): 554-560.

A core concept of infectious disease epidemiology is the abundance threshold, below which an infection is unable to invade or persist. There have been contrasting theoretical predictions regarding the nature of this threshold for vector-borne diseases, but for infections with an invertebrate vector, it is common to assume a threshold defined by the ratio of vector and host abundances. Here, we show in contrast, both from field data and model simulations, that for plague (*Yersinia pestis*) in Kazakhstan, the invasion threshold quantity is based on the product of its host (*Rhombomys opimus*) and vector (mainly *Xenopsylla spp.*) abundances, resulting in a combined threshold curve with hyperbolic shape. This shape implies compensation between host and vector abundances in permitting infection, which has important implications for disease

control. Realistic joint thresholds, supported by data, should promote improved understanding, prediction and management of disease occurrence in this and other vector-borne disease systems.

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Su, H.-Y. , Bai, X.-L. 2012. Monitoring results and epidemic analysis of gerbil plague in northern Ningxia in 2009. Chinese Journal of Endemiology 31 (1) , pp. 91-93.

Objective: To analyze gerbil plague in northern Ningxia and the monitoring results, to master the plague epidemic dynamics, and to provide the basis for developing countermeasures. Methods: Monitoring data of gerbil plague focus in northern Ningxia were collect in 2009, counted main host density, rate of dye fleas, flea body index and bacteriology, serology detect and analyzed the epidemic situation. Results: An average density of main host was 1.74/hm², the average rate of infected fleas was 28.60%, and the average rat body flea index was 0.76. Monitoring found 4 plagues from rat plague epidemic, plague bacteria were found in 16 strains, of which gerbils inspection bacteria 10 strains, *Meriones unguiculatus* 2 strains, the same type *cheopsis* 2 strains, and bald disease fleas 2 strains. Indirect hemagglutination (IIIA) was used to test 529 copies of samples, 2 copies were found positive, and hemagglutination-positive rate was 0.38%; eight copies were examined by reverse indirect hemagglutination (RIHA), 7 material were found positive, and hemagglutination-positive rate was 87.50%. Conclusions: In recent years, the disease is active among animals in gerbil plague focus Ningxia. The density of rodents is higher, and local plague epidemic is found in rats. Monitoring efforts should be strengthened and scope of monitoring should be expanded. We should pay close attention to the epidemic dynamics, control the prevalence and spread of animal disease, and prevent the occurrence of human plague.

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Urich, S.K. , Chalcraft, L., Schriefer, M.E., Yockey, B.M., Petersen, J.M. 2012. Lack of antimicrobial resistance in *Yersinia pestis* isolates from 17 countries in the Americas, Africa, and Asia. Antimicrobial Agents and Chemotherapy 56 (1) , pp. 555-558.

Yersinia pestis is the causative agent of plague, a fulminant disease that is often fatal without antimicrobial treatment. Plasmid (IncA/C)-mediated multidrug resistance in *Y. pestis* was reported in 1995 in Madagascar and has generated considerable public health concern, most recently because of the identification of IncA/C multidrug-resistant plasmids in other zoonotic pathogens. Here, we demonstrate no resistance in 392 *Y. pestis* isolates from 17 countries to eight antimicrobials used for treatment or prophylaxis of plague.

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Voegeli, B; Saladin, V; Wegmann, M; Richner, H; AF Voegeli, B.; Saladin, V.; Wegmann, M.; Richner, H. 2012. Parasites as mediators of heterozygosity-fitness correlations in the Great Tit (*Parus major*). JOURNAL OF EVOLUTIONARY BIOLOGY 25 (3): 584- 590.

Positive correlations between heterozygosity and fitness traits are frequently observed, and it has been hypothesized, but rarely tested experimentally, that parasites play a key role in mediating the heterozygosity-fitness association. We evaluated this hypothesis in a wild great tit (*Parus major*) population by testing the prediction that the heterozygosity-fitness association would appear in broods experimentally infested with a common ectoparasite, but not in parasite-free broods. We simultaneously assessed the effects of parental and offspring heterozygosity on nestling growth and found that body mass of nestlings close to independence, which is a strong predictor of post-fledging survival, increased significantly with nestling levels of heterozygosity in experimentally infested nests, but not in parasite-free nests. Heterozygosity level of the fathers also showed a significant positive correlation with offspring body mass under an experimental parasite load, whereas there was no correlation with the mothers level of heterozygosity. Thus, our results indicate a key role for parasites as mediators of the heterozygosity-fitness correlations.

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Von Stein, R.T. , Soderlund, D.M. 2012. Role of the local anesthetic receptor in the state-dependent inhibition of voltage-gated sodium channels by the insecticide metaflumizone. *Molecular Pharmacology* 81 (3) , pp. 366-374.

Sodium channel inhibitor (SCI) insecticides selectively target voltage-gated sodium (Na v) channels in the slow-inactivated state by *binding* at or near the local anesthetic receptor within the sodium channel pore. Metaflumizone is a new insecticide for the treatment of fleas on domesticated pets and has recently been reported to block insect sodium channels in the slow-inactivated state, thereby implying that it is also a member of the SCI class. Using the two-electrode voltage-clamp technique, we examined metaflumizone inhibition of rat Na v1.4 sodium channels expressed in *Xenopus laevis* oocytes. Metaflumizone selectively inhibited Na v1.4 channels at potentials that promoted slow inactivation and shifted the voltage dependence of slow inactivation in the direction of hyperpolarization. Metaflumizone perfusion at a hyperpolarized holding potential also shifted the conductance-voltage curve for activation in the direction of depolarization and antagonized use-dependent lidocaine inhibition of fast-inactivated sodium channels, actions not previously observed with other SCI insecticides. We expressed mutated Na v1.4/F1579A and Na v1.4/Y1586A channels to investigate whether metaflumizone shares the domain IV segment S6 (DIV-S6) binding determinants identified for other SCI insecticides. Consistent with previous investigations of SCI insecticides on rat Na v1.4 channels, the F1579A mutation reduced sensitivity to block by metaflumizone, whereas the Y1586A mutation paradoxically increased the sensitivity to metaflumizone. We conclude that metaflumizone selectively inhibits slow-inactivated Na v1.4 channels and shares DIV-S6 binding determinants with other SCI insecticides and therapeutic drugs. However, our results suggest that metaflumizone interacts with resting and fast-inactivated channels in a manner that is distinct from other compounds in this insecticide class.

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Fleas in Art and History

LE PARASITE (COMÉDIE), 1654, by MR. TRISTAN

"SCÈNE PREMIÈRE.

PHÉNICE:

Que le poste est mauvais pour une confidente,
De passer une nuit près d'une jeune Amante !
Elle est à babiller du soir jusqu'au matin,
Et l'on dormirait mieux près de quelque Lutin.

5 Ô l'importun effet d'une amoureuse cause !

L'on dit et l'on redit cent fois la même chose,
On se souvient de tout, et l'on en vient troubler
Celles qui du sommeil se sentent accabler.
Que de propos divers dessus une vétille ?

10 On soupire sans cesse, à toute heure on frétille ;
On vient vous demander, en vous tirant le bras,
Dites-moi, dormez-vous ? Ou ne dormez-vous pas ?
Lucinde sans mentir, n'a point de conscience :

Elle ne m'a donné, ni paix, ni patience,

15 J'en aurai ce matin les yeux tous endormis :
J'aimerais mieux coucher près d'un tas de fourmis.
Cent puces dans mon lit m'auraient moins éveillée ;
Mais la voici venir. Quoi ? Si tôt habillée ?
Déjà sur mes talons ? Quoi donc ? "

[FIRST SCENE.

PHENICE:

For the position that is bad for a confidante,
Spend a night beside a young Lover!
She is babbling from evening to morning,
And we sleep better near some Imp.

5 O the importuning effect of a loving cause!
It is said and repeated a hundred times the same thing,
It remembers everything, and it invents trouble
Those who sleep feel overwhelmed.
How about a trifle over-different?

10 One sighs constantly, at any hour one frets;
One just asks you, pulling your arm,
Tell me, do you sleep? Or do you not sleep?
Lucinde, without lying, has no conscience;
She gave me no peace, no patience,
15 I will have this morning, eyes all asleep:
I would rather sleep near a stash of ants.

A hundred fleas in my bed and I would be less awake;
 But here she comes. What? Dressed so early?
 Already on my heels? What, then?]

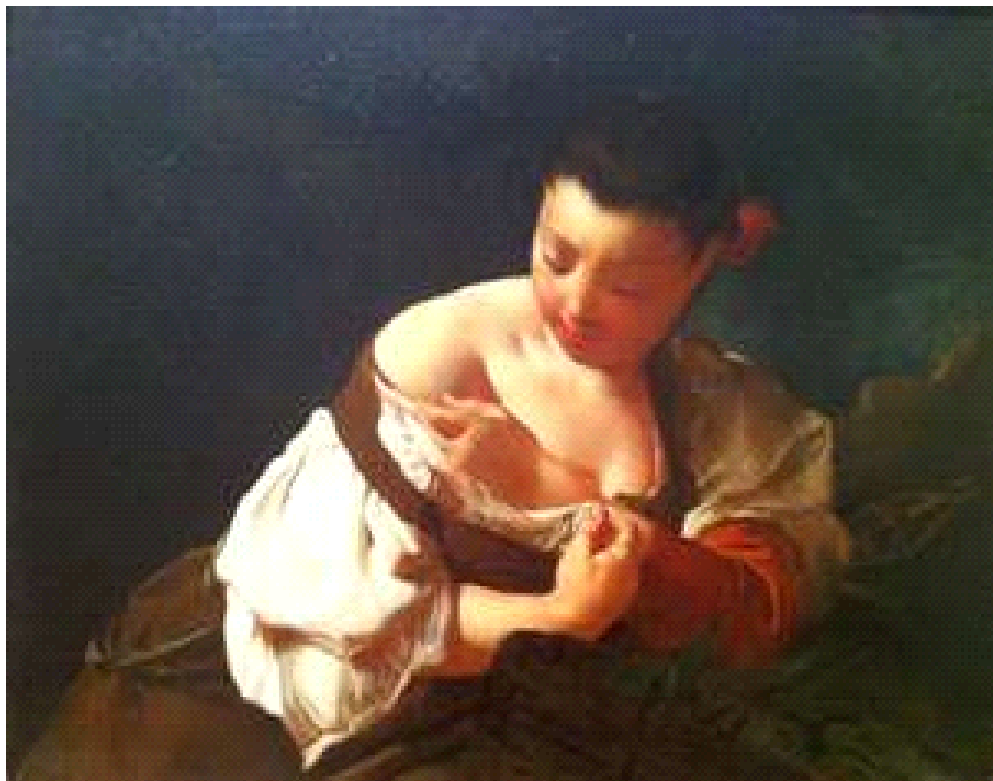
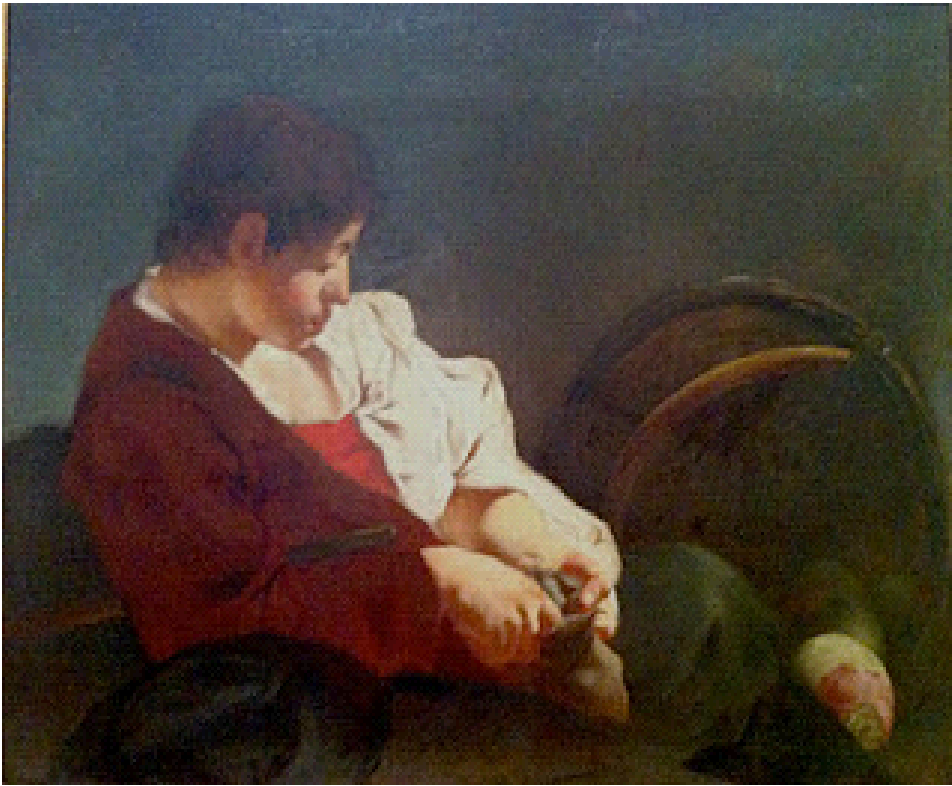
"This essay argues that Tristan L'Hermite's comedy, Le Parasite (1653-54), uses the figure of the parasite to reconceptualize a central tension of Baroque theater: the problematic relationship of appearances to reality, or what William Egginton has called its problem of truth. By putting the play's depiction of parasitism into dialogue with more recent explorations of the parasite by Michel Serres and Jacques Derrida, this reading aims to elucidate Tristan's highly original exploration of the nature of authority, the power of mediation, and the role of theatrical language in human relations."

Welch, E.R. 2012. Of flatterers and fleas: Tristan L'Hermite's Le Parasite and Baroque theater's problem of truth. *Symposium - Quarterly Journal in Modern Literatures* 66 (1) , pp. 31-40.

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Next page: Peasant Boy at a Market (top), Peasant Girl Catching a Flea (bottom).
 c. 1715, by Giovanni Battista Piazzetta (1680-1750).

"When, shortly after his return to Venice, Piazzetta ... may have had one of Crespi's numerous representations of Flea Hunts in the back of his mind... unless the two artists drew independently on one of the many Dutch or French prototypes of the 17th century. Girls catching and squeezing fleas, a reference to mercenary love, were popular in Northern Baroque art; solemn Georges de La Tour painted at least one such scene." - Pg. 169, Jane Martineau, and Andrew Robison, (1994). *The Glory of Venice: Art in the Eighteenth Century*. Yale University Press. ISBN 0300061862, 9780300061864, 528 pages.



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"Meanwhile life was hard. The winter was as cold as the last one had been, and food was even shorter. Once again all rations were reduced, except those of the pigs and the dogs. A too rigid equality in rations, Squealer explained, would have been contrary to the principles of Animalism. In any case he had no difficulty in proving to the other animals that they were NOT in reality short of food, whatever the appearances might be. For the time being, certainly, it had been found necessary to make a readjustment of rations (Squealer always spoke of it as a "readjustment," never as a "reduction"), but in comparison with the days of Jones, the improvement was enormous. Reading out the figures in a shrill, rapid voice, he proved to them in detail that they had more oats, more hay, more turnips than they had had in Jones's day, that they worked shorter hours, that their drinking water was of better quality, that they lived longer, that a larger proportion of their young ones survived infancy, and that they had more straw in their stalls and suffered less from fleas. The animals believed every word of it. Truth to tell, Jones and all he stood for had almost faded out of their memories. They knew that life nowadays was harsh and bare, that they were often hungry and often cold, and that they were usually working when they were not asleep. But doubtless it had been worse in the old days. They were glad to believe so. Besides, in those days they had been slaves and now they were free, and that made all the difference, as Squealer did not fail to point out. "

George Orwell, *Animal Farm* (1945)

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